

Standard IEEE 802.15.4 e Redes ZigBee

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IEEE 802.15.4

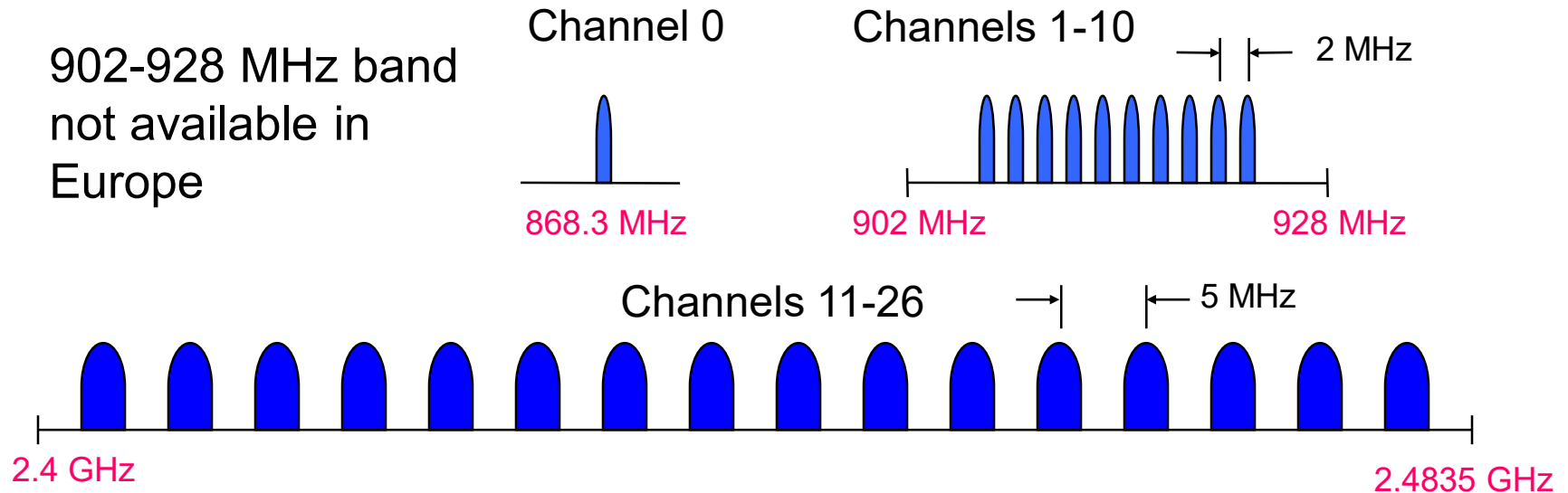
- A standard for wireless sensor networks (PHY and MAC layers)
- Basis of ZigBee, WirelessHART, ISA 100.11a and 6LoWPAN networks
- Data rates up to 250 kbps
- Star or peer-to-peer topologies (other topologies with ZigBee)
- Up to 65535 devices per network
- CSMA/CA channel access, slotted (beacon-enabled) or unslotted
- Support for low-latency devices (beacon-enabled mode, GTS)
- Optional acknowledgement for transfer reliability
- Automatic network establishment by the PAN coordinator
- Dynamic device addressing, flexible addressing format
- Power management to ensure low power consumption
- 16 channels in the 2.4 GHz ISM band, 10 channels in the 915 MHz US ISM band and one channel in the European 868 MHz band

802.15.4 PAN (Personal Area Network)

- Uses a specific 16-bit PAN ID
- A PAN Coordinator “forms” the network
 - Node “0”
 - Chooses radio channel
 - Specifies PAN ID
 - Chooses the security model
 - Starts the network formation process
- Two classes of devices
 - FFD (Full Function Device)
 - Can act as a router
 - Can also perform any other function like sensing
 - RFD (Reduced Function Device)
 - Can’t act as a router, only as an end device
 - Can talk only to a FFD
 - Sensors, actuators, etc. that don’t need to route data

Frequency Bands and Rates

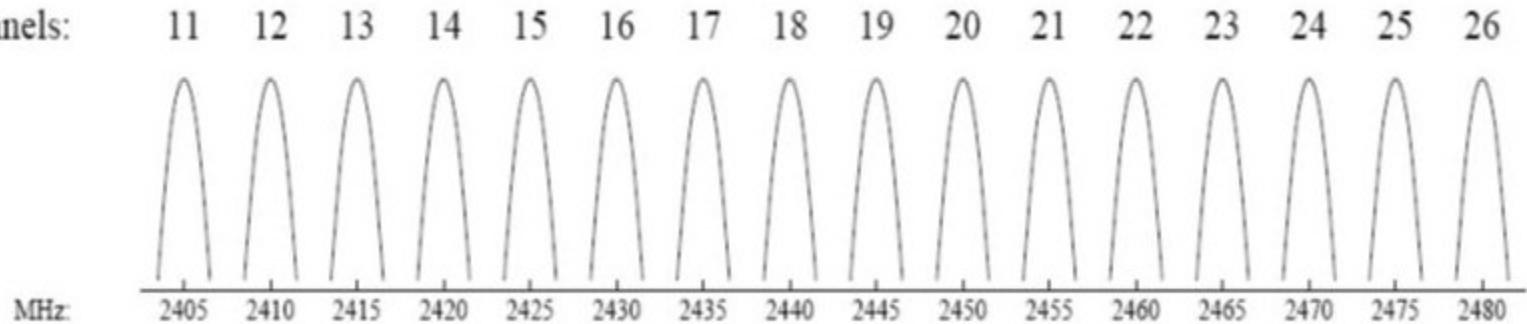
902-928 MHz band
not available in
Europe



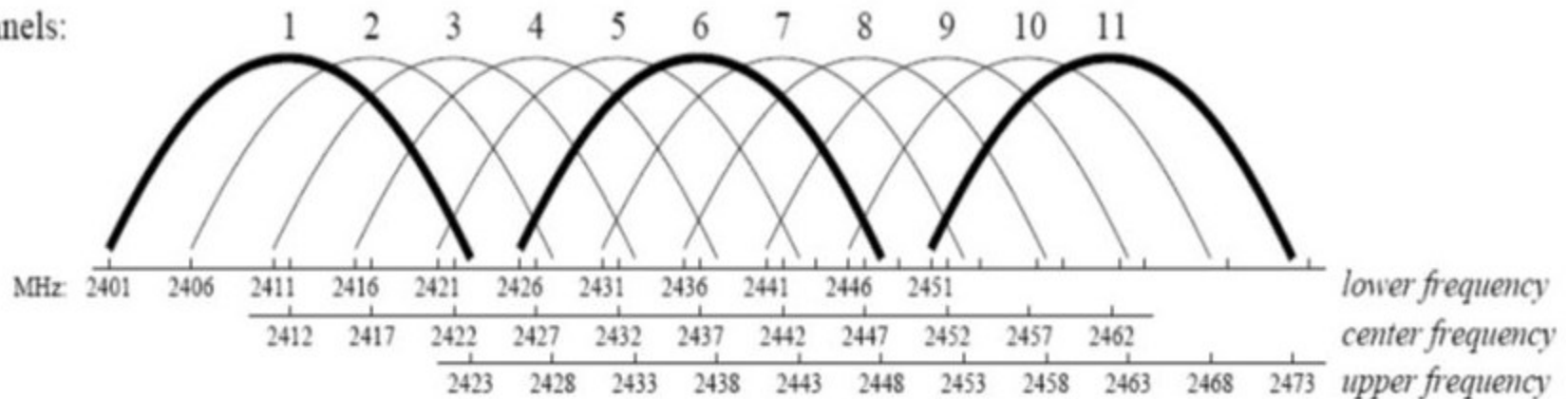
PHY (MHz)	Frequency band (MHz)	Spreading parameters		Data parameters		
		Chip rate (kchip/s)	Modulation	Bit rate (kb/s)	Symbol rate (ksymbol/s)	Symbols
868/915	868-868.6	300	BPSK	20	20	Binary
	902-928	600	BPSK	40	40	Binary
2450	2400-2483.5	2000	O-QPSK	250	62.5	16-ary Orthogonal

Coexistence with Wi-Fi networks

Zigbee Channels:

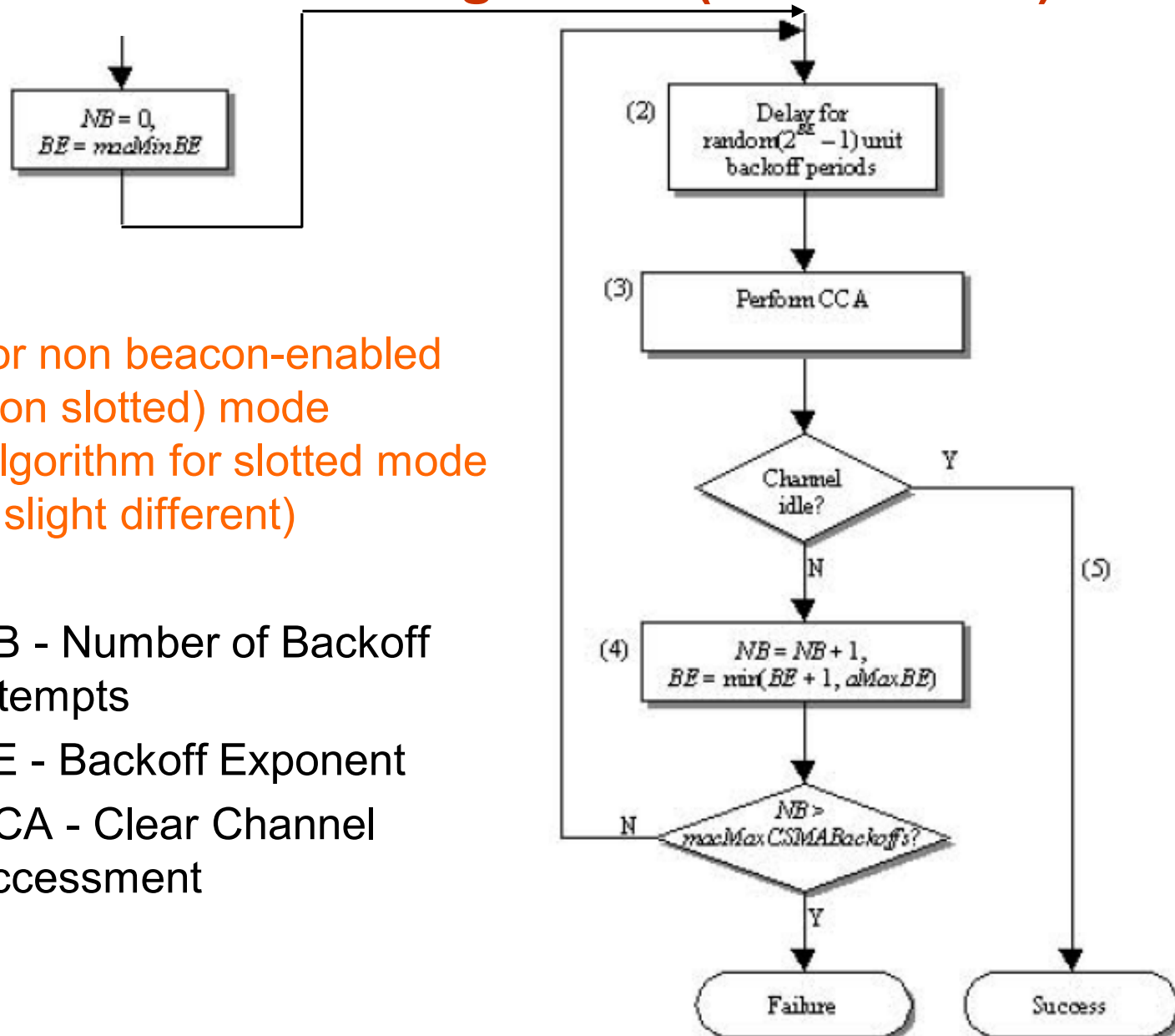


WiFi Channels:

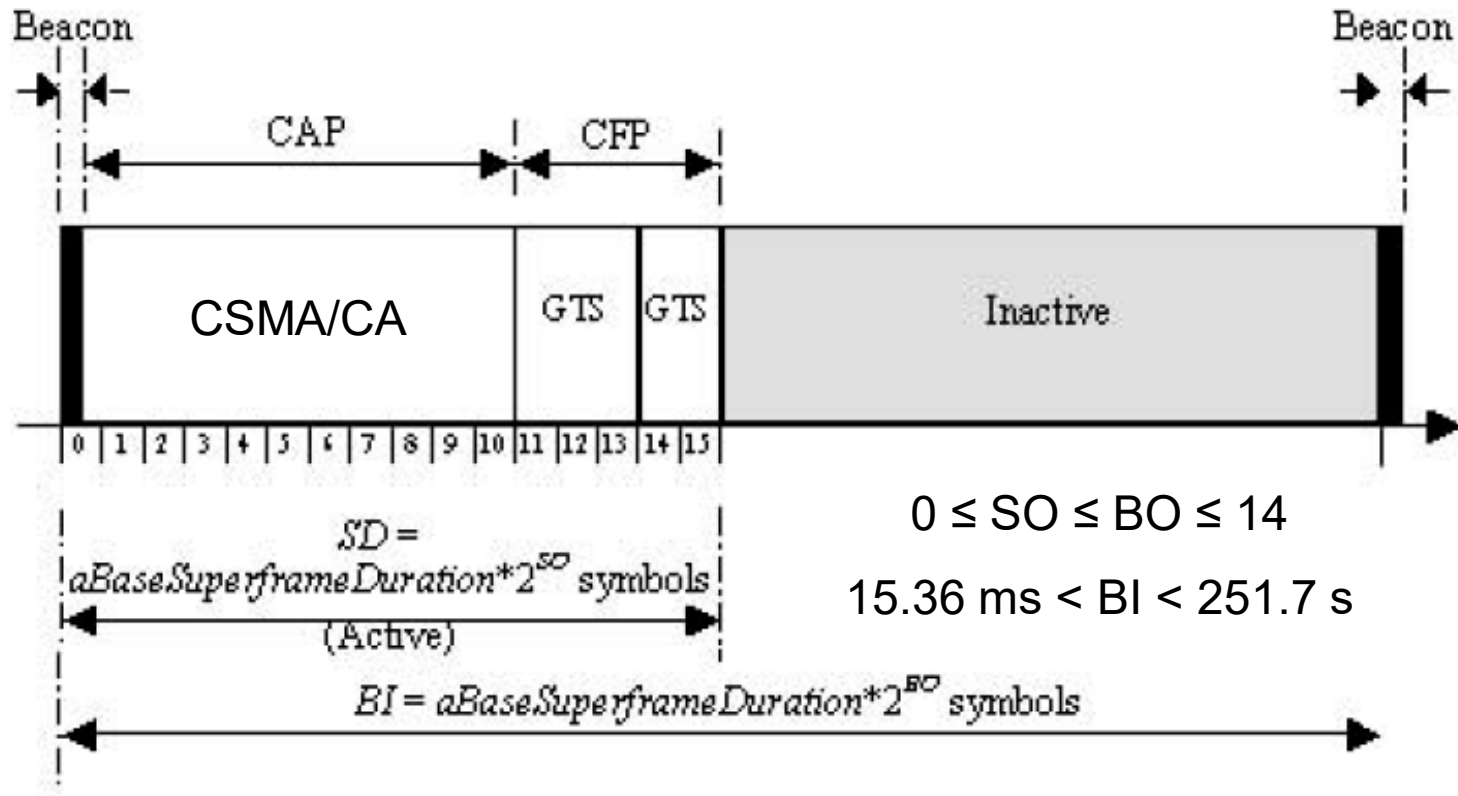


CSMA/CA Algorithm (Non Slotted)

- For non beacon-enabled (non slotted) mode (algorithm for slotted mode is slight different)
- NB - Number of Backoff attempts
- BE - Backoff Exponent
- CCA - Clear Channel Accessment



Superframe Structure (Beacon-enabled mode)



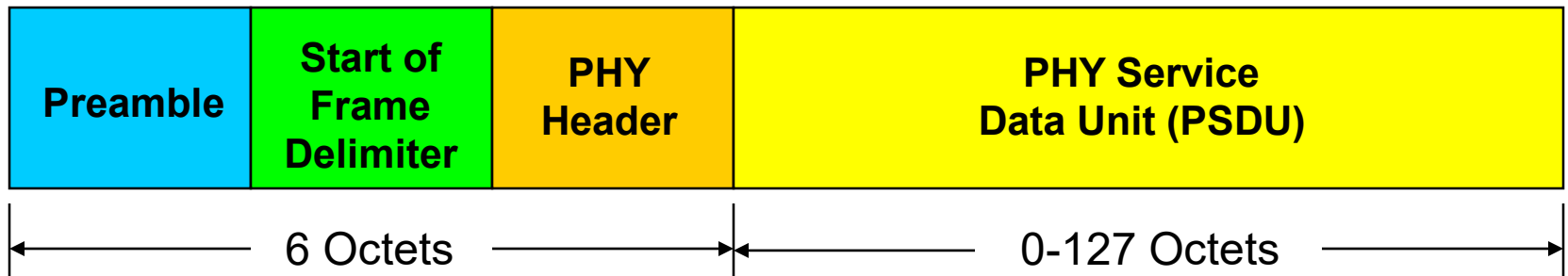
CAP - Contention Access Period
 CFP - Contention Free Period
 GTS - Guaranteed Time Slots

SD - Superframe Duration
 SO - Superframe Order
 BI - Beacon Interval
 BO - Beacon Order

PHY Frame Format

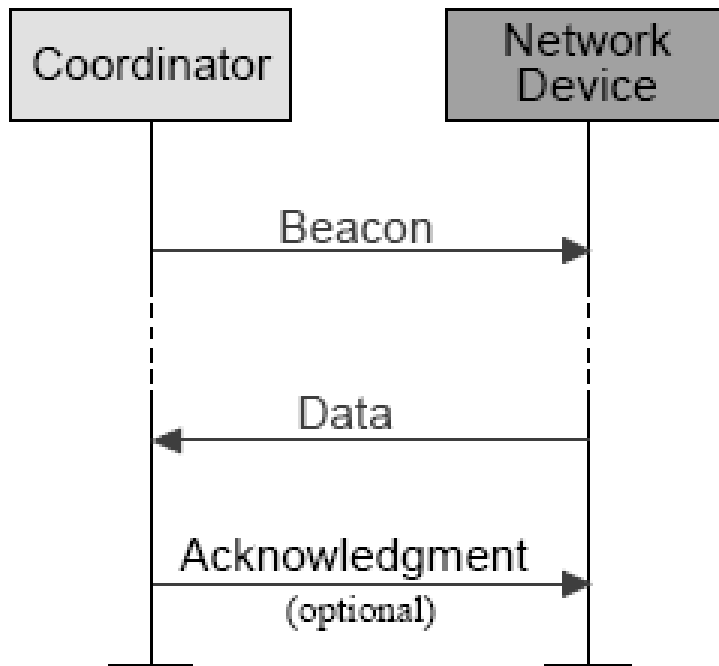
PHY Packet (PPDU) Fields

- **Preamble** (32 bits) – **synchronization**
- Start of Frame Delimiter (**SFD**) (8 bits)
- PHY Header (8 bits) – **PSDU length**
- **PSDU** (0 to 1016 bits) – payload

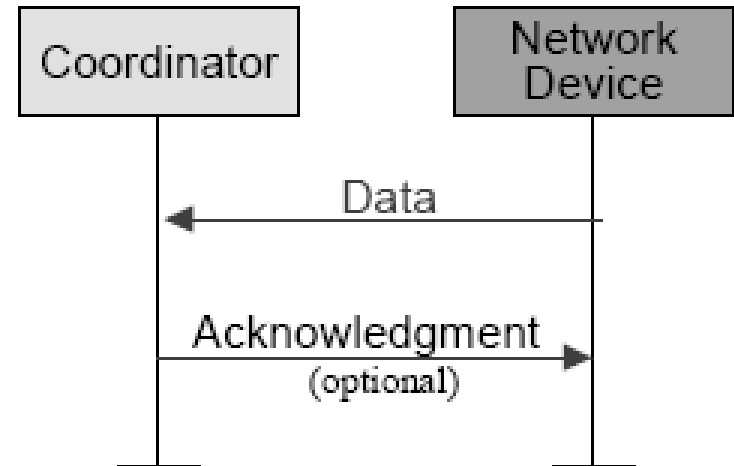


Data Transfer to a Coordinator

Beacon-enabled network

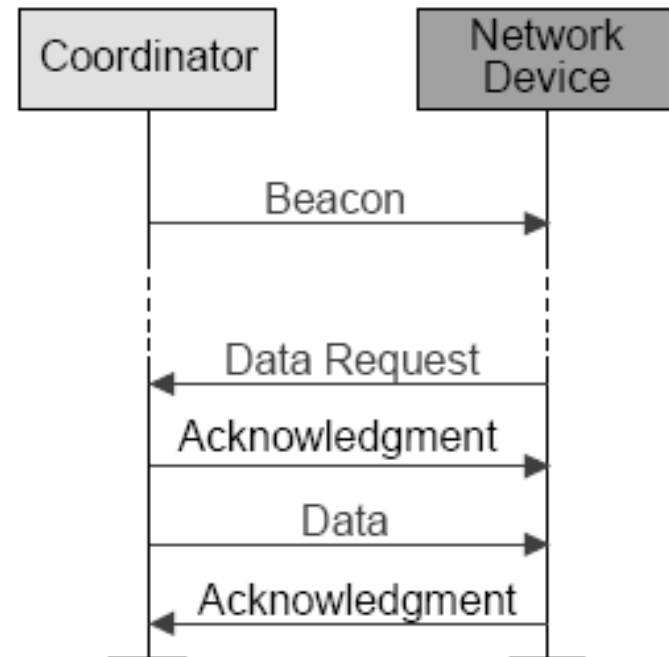


Non beacon-enabled network



Data Transfer from a Coordinator

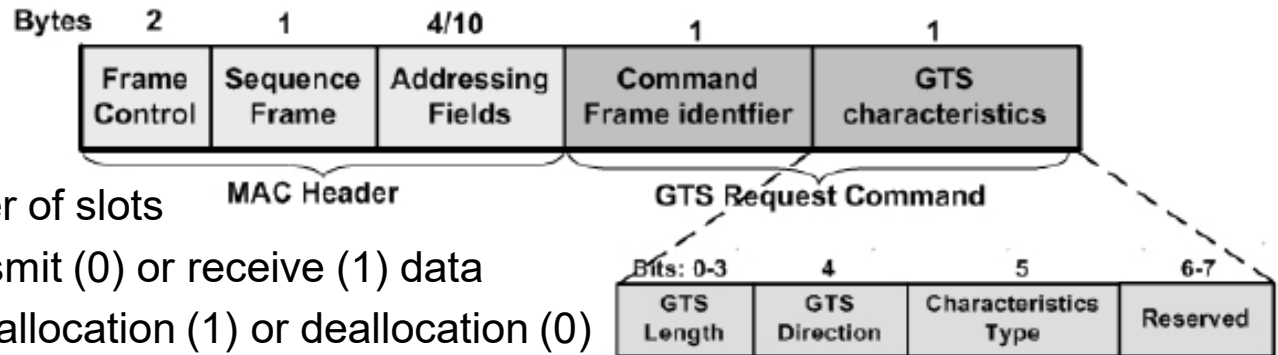
- In a beacon-enabled network:
 - The **coordinator indicates** in the network beacon **that the data message is pending**
 - The **device** periodically wakes up and **listens to the network beacon**
 - If a message is pending, the **device requests the data**, using slotted CSMA-CA
- In a non beacon-enabled network:
 - The **coordinator stores the data and waits** the appropriate device to make a **Data Request**.
 - **If data are not pending, the coordinator transmits a data frame with zero-length payload.**



Guaranteed Time Slot (GTS)

- Allows reservation of slots in a superframe to a given device
- Maximum of 7 GTSs at the same time.

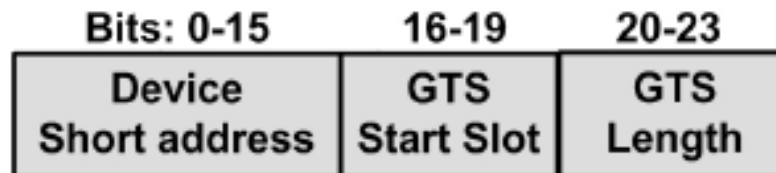
GTS Request



- GTS Length – number of slots
- GTS Direction – transmit (0) or receive (1) data
- Characteristic type – allocation (1) or deallocation (0)

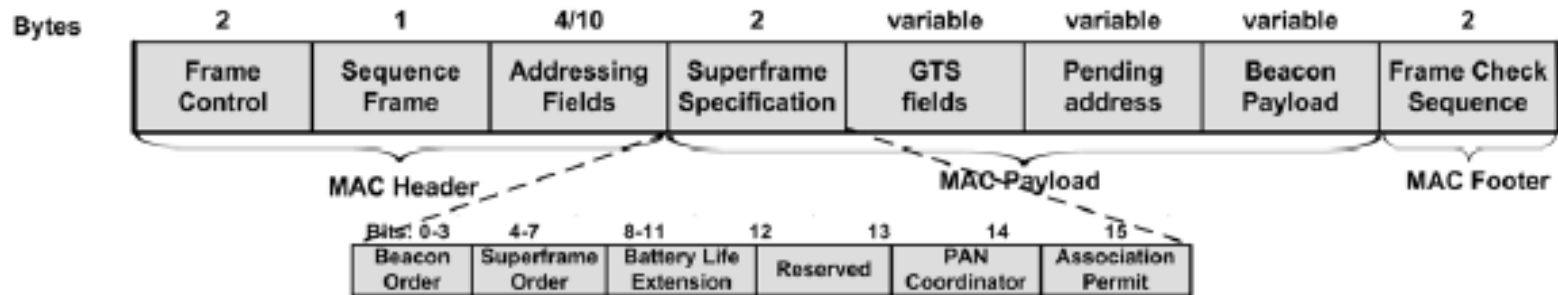
- The result of the GTS request is reported by the coordinator in the beacon frames using a GTS descriptor for each requesting device

GTS Descriptor

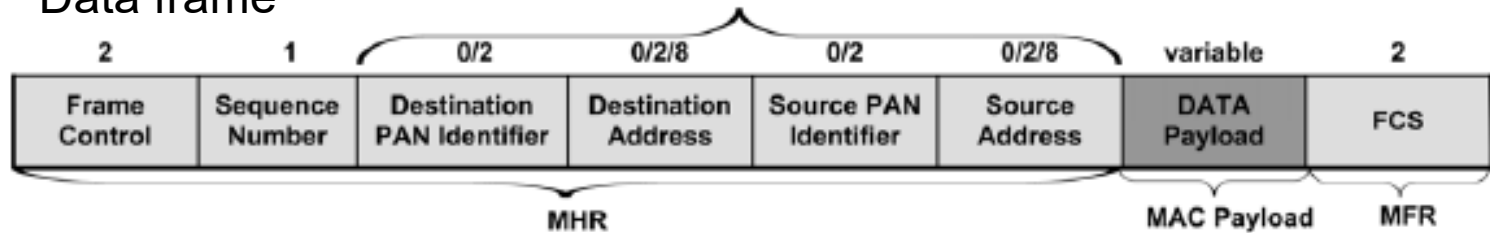


MAC Frame Formats

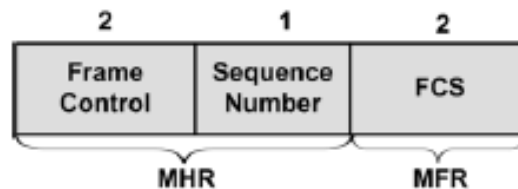
Beacon frame



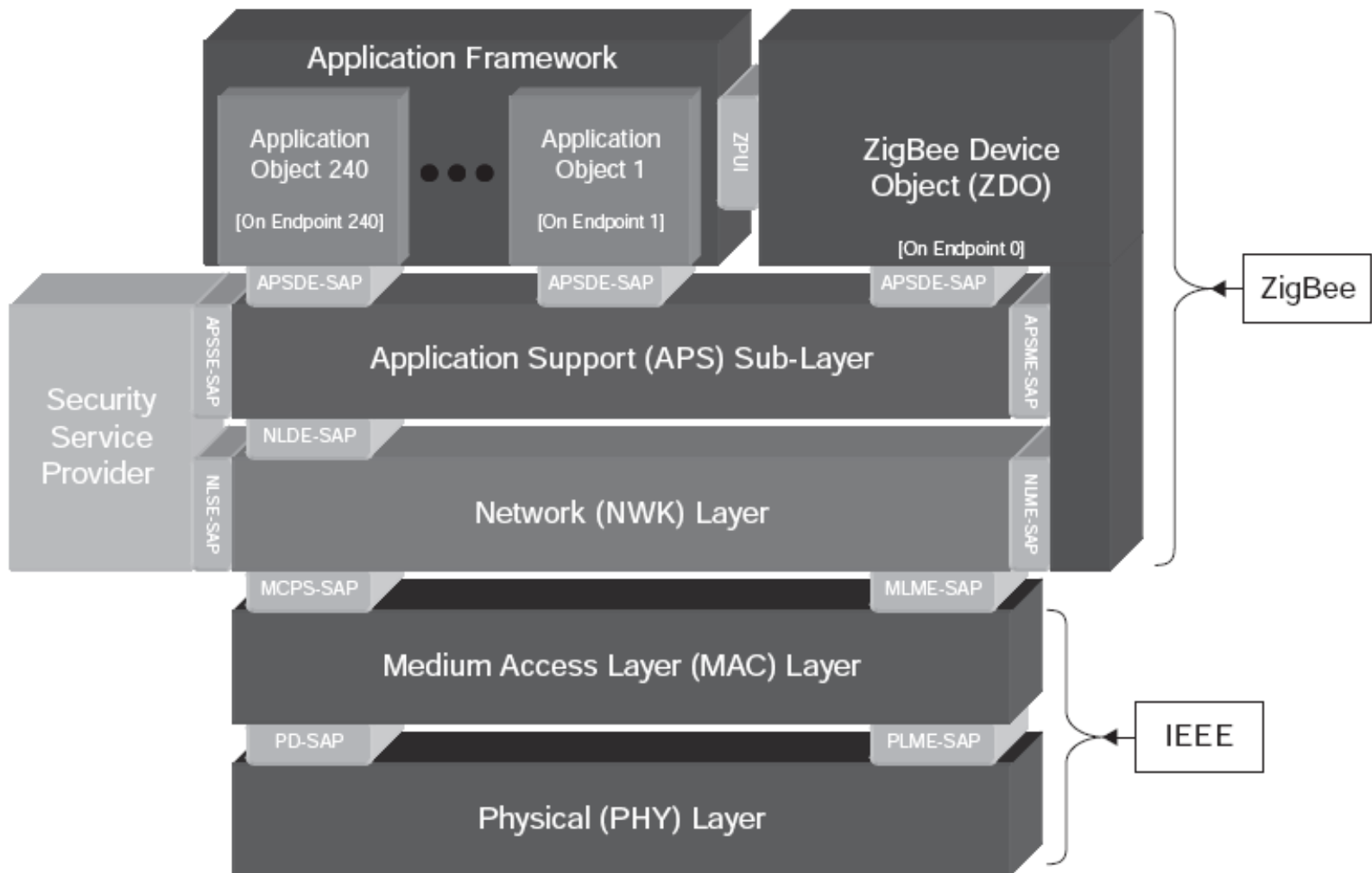
Data frame



ACK frame



ZigBee Stack



ZigBee Node Types

Coordinator (ZC)

- All ZigBee networks must have one (and only one) Coordinator.
 - In the Star topology, the Coordinator is the central node.
 - In the Tree and Mesh topologies, the Coordinator is the top (root) node in the network.

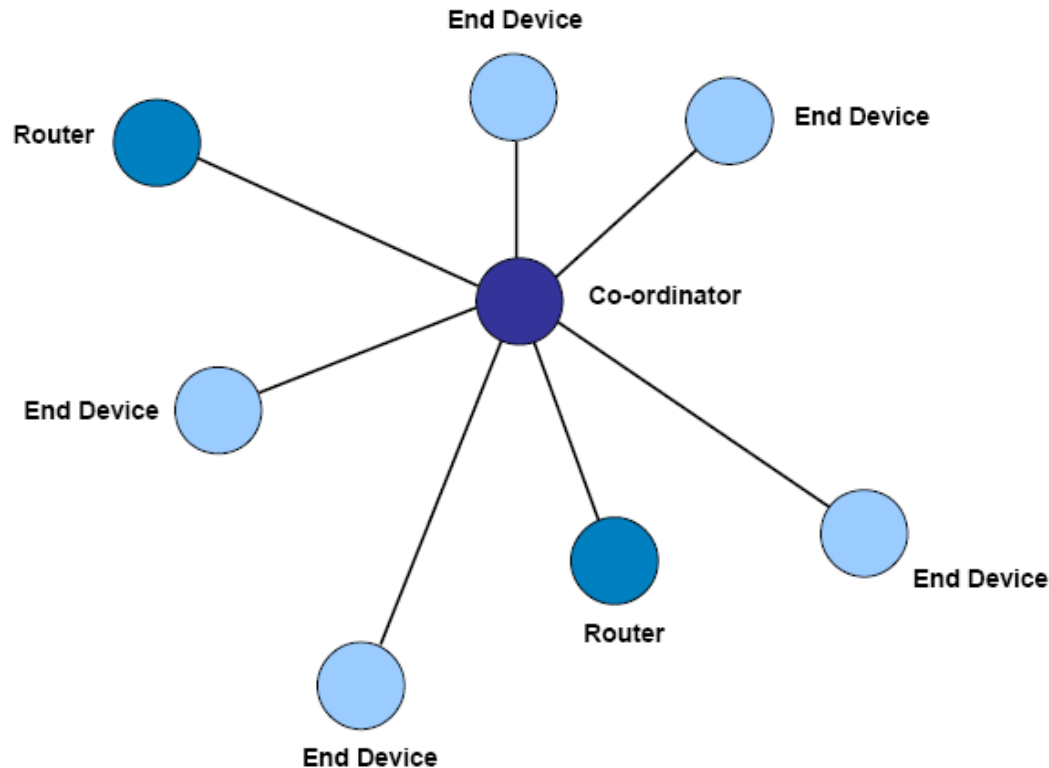
Router (ZR)

- Networks with Tree or Mesh topologies need at least one Router.
- The main tasks of a Router are:
 - Relays messages from one node to another.
 - Allow child nodes to connect to it.
- A Router cannot sleep.

End Device (ZED)

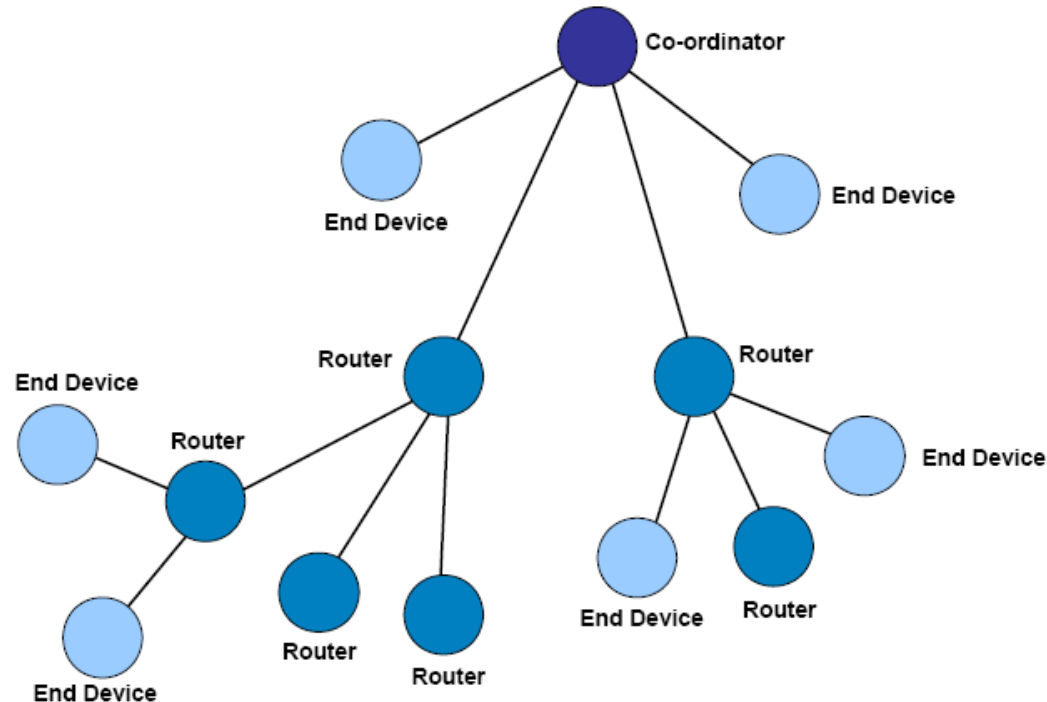
- End Devices are always located at the extremities of a network.
- An End Device can often be battery-powered and, when not transmitting or receiving, can sleep in order to conserve power.

Star Topology



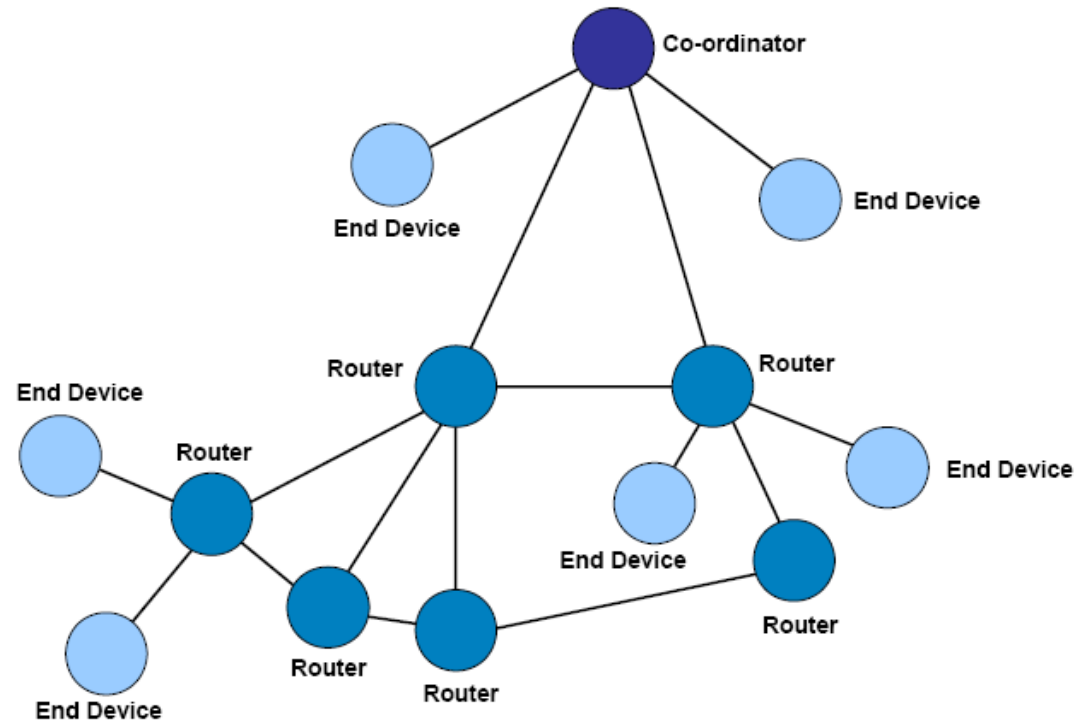
- The network has a central node (coordinator), which is linked to all other nodes in the network.
- All messages travel through the central node.

Tree Topology



- The network has a top node with a branch/leaf structure below.
- To reach its destination, a message travels up the tree (as far as necessary) and then down the tree.
- A disadvantage of this topology is that there is **no alternative route** if a necessary link fails.

Mesh Topology



- Router nodes within range of each other can communicate directly.
- Alternative routes can be found if a link fails or there is congestion.

ZigBee Coordinator Duties

- Starts the network formation.
- Selects the 802.15.4 channel on which the network will operate.
- Selects the extended and short PAN ID for the network.
- Decides on the stack profile to use (compile or run-time option).
- Acts as the Trust Center, storing security keys and using security services to authorize a device onto the network.
- Acts as a router for mesh routing.

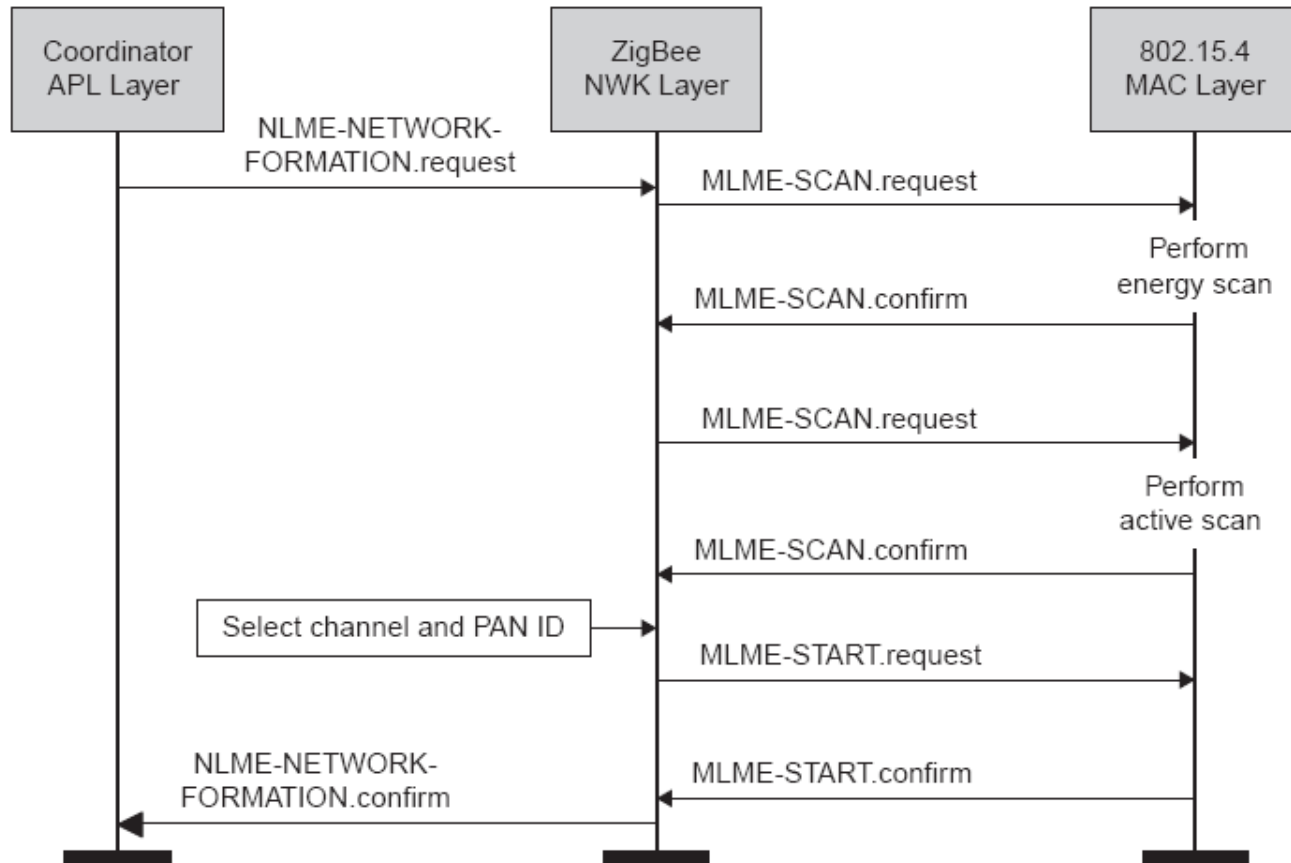
ZigBee Router Duties

- Finding and joining the “correct” network.
- Perpetuating broadcasts across the network.
- Participating in routing, including discovering and maintaining routes.
- Allowing other devices to join the network (if permit-join enabled).
- Storing packets addressed to sleeping children.

ZigBee End Device Duties

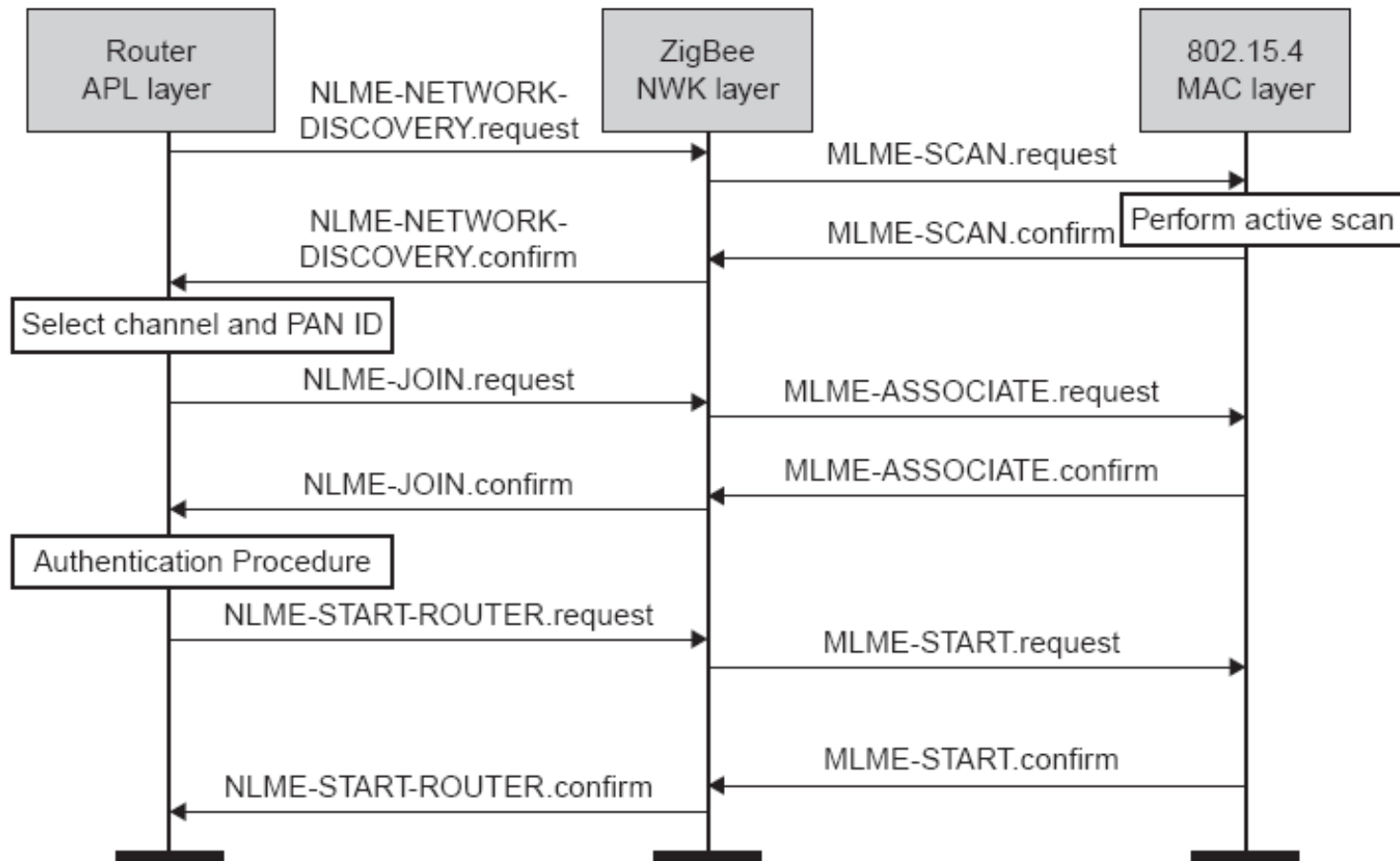
- Finding and joining the “correct” network.
- Polling their parents (data request) to see if any messages were sent to them while they were asleep.
- Finding a new parent if the link to the old parent is lost (NWK rejoin).
- Sleeping most of the time to conserve battery energy when not in use by the application.

ZC Network Formation



- **Active scan** - a MAC beacon request followed by zero or more beacon responses is used to discover what other networks are in the vicinity and ensure the coordinator does not form a network using an existing PAN ID.

ZR Network Joining Process



ZigBee Addressing

- ZigBee uses **two addresses per node**:
 - **long** address (64 bits) – **unique** in the world
 - **short** address (16 bits) – **attributed at network joining**
- Two source and two destination **addressing fields**:
 - **MAC** header – link nodes
 - **NWK** header – end nodes
- Short **address assignment schemes**:
 - **Cskip** - used by **ZigBee** (stack profile 0x01)
 - **Deterministic** assigned **according to the node position** in a symmetrical tree structure
 - Scalability of the network constrained to 10 hops
 - **Stochastic** - used by **ZigBee Pro** (stack profile 0x02)
 - **Randomly** assigned

ZigBee Routing

- Broadcasting
- Mesh routing
 - Based on AODV (Ad hoc On-Demand Distance Vector) routing protocol
 - Requires a routing table on each router
 - The route discovering process is performed on demand
 - A route is kept until it fails or it is removed from the routing table
 - When an existing route fails, the originating node is informed and starts a new route discovery (self-healing)
- Tree routing (only in stack profile 0x01)
 - Based on the Cskip addressing
- Source routing (only in stack profile 0x02)
 - Routes are stored in a single node and sent within the packets
 - Limited to a maximum of 5 hops

Application Profiles

- Collections of common definitions and protocols that allow various devices to work together in a particular domain
- Allow device interoperability across different manufacturers
- Each profile defines device types and required functionality
- Profiles can be Public (developed and maintained by the ZigBee Alliance) or Private (manufacturer specific)
- Each public or private profile has a name and a 16-bit numeric identifier to tag its messages.
 - For example, the Smart Energy profile uses 0x0109.
 - Any device that wants to support the Smart Energy profile is required to provide certain standard functions and to identify its messages with 0x0109 to all other devices.

Application Profiles

- Some public profiles
 - Health Care
 - Home Automation
 - Building Automation
 - Smart Energy
- Examples of use
 - The Smart Energy profile allows different brands of electric meters, thermostats, appliances, and in-home display units to share a common language.
 - Any ZigBee Health Care-certified patient sensor can communicate medical data to any other brand of Health Care-certified patient monitor

Example: Devices of Home Automation Profile

